DocAI OCR Engineering: Machine Learning Error Correction Patterns for MRZ

Type: technical_guide | **Date:** 2023-02-14

1. Common Optical Confusion Pairs

Under low-light or glare conditions, OCR engines frequently confuse optical glyphs: digit '0' and letter 'O', digit '1' and letter 'I' or 'L', digit '5' and letter 'S', digit '8' and letter 'B', digit '2' and letter 'Z'. In MRZ processing, positions are strictly typed (alpha-only vs numeric-only), enabling 100% deterministic pre-correction.

2. Chevrons and Delimiter Recovery

Filler chevrons '<' are frequently misrecognized as '(', '[', '{', or 'K' when printed with low ink density. A specialized regex pass converts non-alphanumeric punctuation inside MRZ boundaries back to '<'. Any line with length differing from 30 (TD1) or 44 (TD3) must trigger boundary recalibration.